# Junkai Zhang

CS PhD Candidate @ UCLA

Website / Google Scholar / Semantic Scholar / LinkedIn / jkzhang@g.ucla.edu

## Education

**University of California, Los Angeles** — Sep. 2022 – Jun. 2027 (Expected)
Ph.D. in Computer Science, advisor: Professor Wei Wang — Los Angeles, CA, US

**University of California, Los Angeles** — Sep. 2022 – Jun. 2025
M.S. in Computer Science — Los Angeles, CA, US

**Tsinghua University** — Aug. 2018 – Jun. 2022
B.S. in Mathematics, Minor in Philosophy — Beijing, CN

## Research Interests

**Reinforcement Learning, Agents, AI for Science**

## Experience

**Point72 Asset Management** — Jun 2026 – Sep 2026
Quantitative Researcher Intern — New York, NY

**Google** — Oct 2025 – Dec 2025
Student Researcher — Los Angeles, CA

**Scale AI** — Jun 2025 – Sep 2025
Research Intern, Post-training — San Francisco, CA

**University of California, Los Angeles** — Sep 2022 – Present
Graduate Student Researcher / Teaching Assistant — Los Angeles, CA

## Publications

The asterisk * indicates equal contributions.

SELECTED WORKS

**Thinking with Visual Grounding**
**Junkai Zhang**, Yihe Deng, Kai-Wei Chang, Wei Wang
in *EMNLP 2026 Findings*

**Chasing the Tail: Effective Rubric-based Reward Modeling for Large Language Model Post-Training**
**Junkai Zhang***, Zihao Wang*, Lin Gui*, Swarnashree Mysore Sathyendra, Jaehwan Jeong, Victor Veitch, Wei Wang, Yunzhong He, Bing Liu, Lifeng Jin
in *ICLR 2026*

### Enhancing LLM Safety Through a Theoretical Minimax Game Lens

Yihe Deng*, Yu Yang*, **Junkai Zhang***, Wei Wang, Bo Li

in *AISTATS 2026*

## REINFORCEMENT LEARNING

### Uncertainty-Aware Reward-Free Exploration with General Function Approximation

**Junkai Zhang***, Weitong Zhang*, Dongruo Zhou, Quanquan Gu

in *ICML 2024*

### Optimal Horizon-Free Reward-Free Exploration for Linear Mixture MDPs

**Junkai Zhang**, Weitong Zhang, Quanquan Gu

in *ICML 2023*

## OPTIMIZATION METHODS

### Why Does Sharpness-Aware Minimization Generalize Better Than SGD?

Zixiang Chen*, **Junkai Zhang***, Yiwen Kou, Xiangning Chen, Cho-Jui Hsieh, Quanquan Gu

in *NeurIPS 2023*

## AI FOR SCIENCE

### MatSciBench: Benchmarking the Reasoning Ability of LLM in Material Science

**Junkai Zhang***, Jingru Gan*, Zian Jia, Changquan Gu, Jianpeng Chen, Xiaoxuan Wang, Yanqiao Zhu, Mingyu Derek Ma, Dawei Zhou, Ling Li, Wei Wang

in *KDD 2026*

### METASYMBO: Multi-Agent Language-Guided Metamaterial Discovery via Symbolic Latent Evolution

Jianpeng Chen, Wangzhi Zhan, Dongqi Fu, **Junkai Zhang**, Zian Jia, Ling Li, Wei Wang, Dawei Zhou

in *COLM 2026*

### MetaScientist: A Human-AI Synergistic Framework for Automated Mechanical Metamaterial Design

Jingyuan Qi, Zian Jia, Minqian Liu, Wangzhi Zhan, **Junkai Zhang**, Xiaofei Wen, Jingru Gan, Jianpeng Chen, Qin Liu, Mingyu Derek Ma, Bangzheng Li, Haohui Wang, Adithya Kulkarni, Muhao Chen, Dawei Zhou, Ling Li, Wei Wang, Lifu Huang

in *NAACL 2025 Demo*

### Protein Large Language Models: A Comprehensive Survey

Yijia Xiao, Wanjia Zhao, **Junkai Zhang**, Yiqiao Jin, Han Zhang, Zhicheng Ren, Renliang Sun, Haixin Wang, Guancheng Wan, Pan Lu, Xiao Luo, Yu Zhang, James Zou, Yizhou Sun, Wei Wang

in *EMNLP 2025 Findings*

### MetamatBench: Integrating Heterogeneous Data, Computational Tools, and Visual Interface for Metamaterial Discovery

Jianpeng Chen, Wangzhi Zhan, Haohui Wang, Zian Jia, Jingru Gan, **Junkai Zhang**, Jingyuan Qi, Tingwei Chen, Lifu Huang, Muhao Chen, Ling Li, Wei Wang, Dawei Zhou

in *KDD 2025 Datasets and Benchmarks Track*

### Neural network-assisted personalized handwriting analysis for Parkinson's disease diagnostics

Guorui Chen, Trinny Tat, Yihao Zhou, Zhaoqi Duan, **Junkai Zhang**, Kamryn Scott, Xun Zhao, Zeyang Liu, Wei Wang, Song Li, Katy A. Cross, Jun Chen

in *Nature Chemical Engineering*

**Self-powered in-stent restenosis diagnosis via magnetoelastic stents**

Guorui Chen, Wi Jin Kim, Youcheng Yang, Yan-Ruide Li, Jing Tian, **Junkai Zhang**, Xun Zhao, Kamryn Scott, Lily G. Defelice, Zeyang Liu, Jing Xu, Tzuchun Chung, Jarod Carol, Yihao Zhou, Anthony C. Wang, Olujimi A. Ajijola, Paul S. Weiss, Wei Wang, Song Li, Geoffrey P. Colby, Jun Chen

in *Nature Cardiovascular Research*

**Continuous Treatment Effect Modeling in Multi-agent Dynamical Systems**

Zijie Huang*, Jeehyun Hwang*, **Junkai Zhang***, Jinwoo Baik, Weitong Zhang, Quanquan Gu, Dominik Wodarz, Yizhou Sun, Wei Wang

in *WWW 2024*

GENERATIVE MODELS

**Fast Sampling via De-randomization for Discrete Diffusion Models**

Zixiang Chen, Huizhuo Yuan, Yongqian Li, Yiwen Kou, **Junkai Zhang**, Quanquan Gu

in *NeurIPS 2024*

## Academic Services

**Conference Reviewer:** ICML[2025], ICLR[2024-2026], NeurIPS[2025], NAACL[2025], AAAI[2024-2025], AISTATS[2024-2025]